

ANALYSTLAB AFRICA
Experience Lab Internship Programme

FinTrust Digital Bank - Week 2 Project Documentation

Project: FinTrust Financial Intelligence & Digital Banking Support Solution
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A. Application of the Week 1 Plan

What I originally planned to accomplish

In Week 1 I defined the predictive problem - whether transaction and customer information can help predict the Risk_Review_Flag, assessed the target, listed candidate features, and outlined a plan covering data preparation, exploratory analysis, feature engineering and a baseline model.

What I implemented in Week 2

- Loaded, joined, and quality-checked the FinTrust customer and transaction data.
- Carried out a full data-quality assessment covering missing values, duplicates, data types, unusual values and the relationship between the two datasets.
- Produced exploratory data analysis (9 visualizations) covering transaction amount, type, channel, device type, location, international status, transaction status, customer segment, account type and digital engagement against the risk-review flag.
- Engineered 8 features documented with how each was created and why it may matter.
- Built a baseline Logistic Regression classification model, trained and evaluated it on a held-out test set.
- Interpreted the model's coefficients and documented its limitations.

Changes to my approach and why

The plan was implemented largely as designed. Two small refinements were made once I started working with the real data: missing Device_Type and Location values were filled with "Unknown" instead of being dropped, so no transaction records were lost; and numeric features were scaled before training the Logistic Regression model so that the large-magnitude transaction amount did not dominate the model purely because of its scale.

B. Use of the Approved Project Resources

Resource	How it was used	What it provided	Limitation encountered
FinTrust_Customer_Data.csv	Joined to the transaction data on Customer_ID to add customer attributes	1,500 customer records with 12 attributes (age, segment, tenure, engagement, etc.)	No missing values or duplicates found
FinTrust_Transaction_Data.csv	Main analysis and modelling dataset - one row per transaction	12,000 transaction records with 11 attributes, including the Risk_Review_Flag target	Device_Type and Location had a small number of missing values
FinTrust_Data_Dictionary	Used to check field definitions and which variables were intended for modelling	Definitions and modelling guidance for the customer fields	Only covers the customer fields, not the transaction fields
FinTrust Financial Knowledge Base / Project Brief / Roadmap	Used for overall project context and the reminder that Risk_Review_Flag is synthetic	Business framing and scope for the Data Science track	Not directly used for the coding deliverables

C. Tools Used

- Python — for all data preparation, analysis and modelling
- Jupyter Notebook — coding environment for both deliverables
- Pandas / NumPy — data loading, cleaning, joining and feature engineering
- Matplotlib / Seaborn — exploratory visualisations
- Scikit-learn — train/test split, feature scaling, Logistic Regression model, and evaluation metrics

These are the tools specified for the Data Science track in the Week 2 assignment brief.

D. Practical Outputs Produced

- Prepared Modelling Dataset - FinTrust_Modelling_Dataset_Prepared.csv
- EDA Notebook
- Feature Engineering documentation
- Baseline Model - Logistic Regression
- Model Evaluation - Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix and ROC curve
- Model Interpretation - coefficient chart and Feature → Evidence → Interpretation table
- This Week 2 Project Documentation

E. Testing / Evaluation of the Work

Work Completed	Test / Evaluation	Finding	Action / Improvement	Result
Data preparation (join, missing values, data types)	Checked row counts, missing-value counts and duplicate counts after each step	Device_Type and Location had a small number of missing values; no duplicates found	Filled missing values with "Unknown" instead of dropping rows	No transactions lost; 12,000 rows preserved in the prepared dataset
Baseline Logistic Regression model	Evaluated on a held-out 20% test set using Accuracy, Precision, Recall, F1-score and ROC-AUC	Model reached ROC-AUC $\approx$ 0.67, better than random guessing, but with a moderate precision	Applied class_weight="balanced" and scaled numeric features to help the model learn from the imbalanced target	Model reliably identifies a meaningful share of flagged transactions (Recall $\approx$ 59%) while beating the "always No" baseline on ranking ability
Feature engineering	Compared feature values against the EDA findings	Features such as Is_International and Amount_NGN aligned with the patterns seen in the EDA	Kept these features in the model; noted that customer-category features (segment, account type) added little value	Confirmed the features are consistent with the underlying data patterns

F. Important Decisions

Decision	Reason	Evidence	Effect on the Work
Fill missing Device_Type / Location with "Unknown" rather than drop rows	Only a small number of rows were affected	Missing-value count from the data-quality check	No transactions were lost from the dataset
Use Logistic Regression as the baseline model	Simple, fast, and easy to explain to a non-technical audience - a sensible first model	Assignment allows any justified classification algorithm for the baseline	Produced an interpretable model with clear coefficients
Use class_weight="balanced" and a stratified train/test split	The target is imbalanced (~20% flagged)	Target distribution	Prevented the model from simply predicting "No" for every transaction
Scale numeric features before training	Amount_NGN is on a much larger scale than the 0/1 dummy variables	Model failed to converge cleanly without scaling	Model trained properly and produced stable coefficients
Drop Customer_Name and ID columns before modelling	Identifiers carry no predictive information and are not meant for modelling use	Data dictionary guidance and general good practice	Prevented the model from relying on identifiers instead of real patterns

G. Assumptions, Limitations, Risks & Unresolved Issues

Assumptions

- Risk_Review_Flag reflects a consistent, learnable pattern in the data rather than being assigned at random.
- The three months of transaction data provided are representative enough for an initial baseline model.

Data-quality issues

- A small number of missing values in Device_Type and Location (filled with "Unknown").
- Transaction amounts are right-skewed, with a small number of very large transactions.

Technical / methodological limitations

- Logistic Regression is a simple model and may not capture more complex relationships in the data.
- Only one baseline algorithm was tried in this initial round; no hyperparameter tuning was performed.
- Transaction_Status and Account_Status were left out of the model because they may not always be known at the time a transaction takes place.

Risks

- Because Risk_Review_Flag is synthetic, the model's patterns may not transfer to real banking risk indicators.

- Customer_Segment and Account_Type remain in the model despite showing little relationship with the flag; results should be checked across these groups before any further use.

Unresolved issues

- Whether a different baseline algorithm (e.g. Decision Tree or Random Forest) would perform meaningfully better has not yet been tested.
- The effect of including Transaction_Status and Account_Status has not been quantified.

Week 2 Project Summary

1. What I planned to accomplish in Week 2

Prepare and join the FinTrust customer and transaction data, carry out exploratory data analysis against the Risk_Review_Flag target, engineer useful predictive features, and build and evaluate a baseline classification model.

2. What I actually completed

- A full data-quality assessment and a prepared, joined modelling dataset.
- Nine EDA visualizations covering transaction amount, type, channel, device, location, international status, transaction status, customer segment/account type, and digital engagement.
- Eight documented engineered features.
- A baseline Logistic Regression model, evaluated with Accuracy, Precision, Recall, F1-score, ROC-AUC, a confusion matrix and an ROC curve.
- A model interpretation section covering coefficients and a Feature → Evidence → Interpretation table, plus documented limitations.

3. Tools used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

4. Key findings or development outcomes

- About 1 in 5 transactions ($\approx 20\%$) is flagged for review.
- International transactions, transfers and cash withdrawals, and higher transaction amounts are associated with a higher flag rate.
- Channel, device type, location, customer segment, account type and digital engagement showed little relationship with the flag.
- The baseline Logistic Regression model reached a ROC-AUC of about 0.67 and a Recall of about 59% on the held-out test set — a reasonable starting point for an initial model.

5. Major challenges encountered

- Two of the supplied PDF resources were saved as ZIP-packaged files rather than standard PDFs and needed to be unpacked before they could be read.
- The Logistic Regression model initially failed to converge cleanly because of the large difference in scale between transaction amount and the 0/1 category variables; this was resolved by scaling the numeric features.

6. Important decisions made and why

See Section F above for the full decision log. The most significant decisions were to use a class-weighted, scaled Logistic Regression as the baseline model given the imbalanced target, and to fill rather than drop the small number of missing values so no data was lost.

7. Changes to my Week 1 approach

The overall plan from Week 1 was followed closely. The main adjustments were practical ones made once I started working with the real data: filling missing values instead of dropping rows, and scaling numeric features so the model would train properly.

8. Testing or evaluation performed

Data-quality checks (missing values, duplicates, data types) after preparation, and model evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, a confusion matrix and an ROC curve on a held-out test set, as detailed in Section E above.

9. Key limitations

Only one baseline algorithm was tried; and the model has not yet been tested across customer segments for fairness. See Section G for the full list.

10. Remaining work

- Try additional baseline algorithms (e.g. Decision Tree, Random Forest) for comparison.
- Test the effect of including `Transaction_Status` and `Account_Status`.
- Check model predictions across customer segments.

11. Proposed focus for Week 3

Compare the Logistic Regression baseline against at least one additional algorithm, refine the features based on Week 2 findings, and begin a more detailed evaluation of where the model performs well and where it does not.